Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 1.0, **Version Date** 1/14/13

CBF/SAG Alignment Process

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

2 Overview

CBF Alignment ensures the focal spot is accurate, the bowtie filter is centered and center of rotation is in a straight line.

Last Revised:	15 May, 2012

Illustration 1: CBF Flowchart

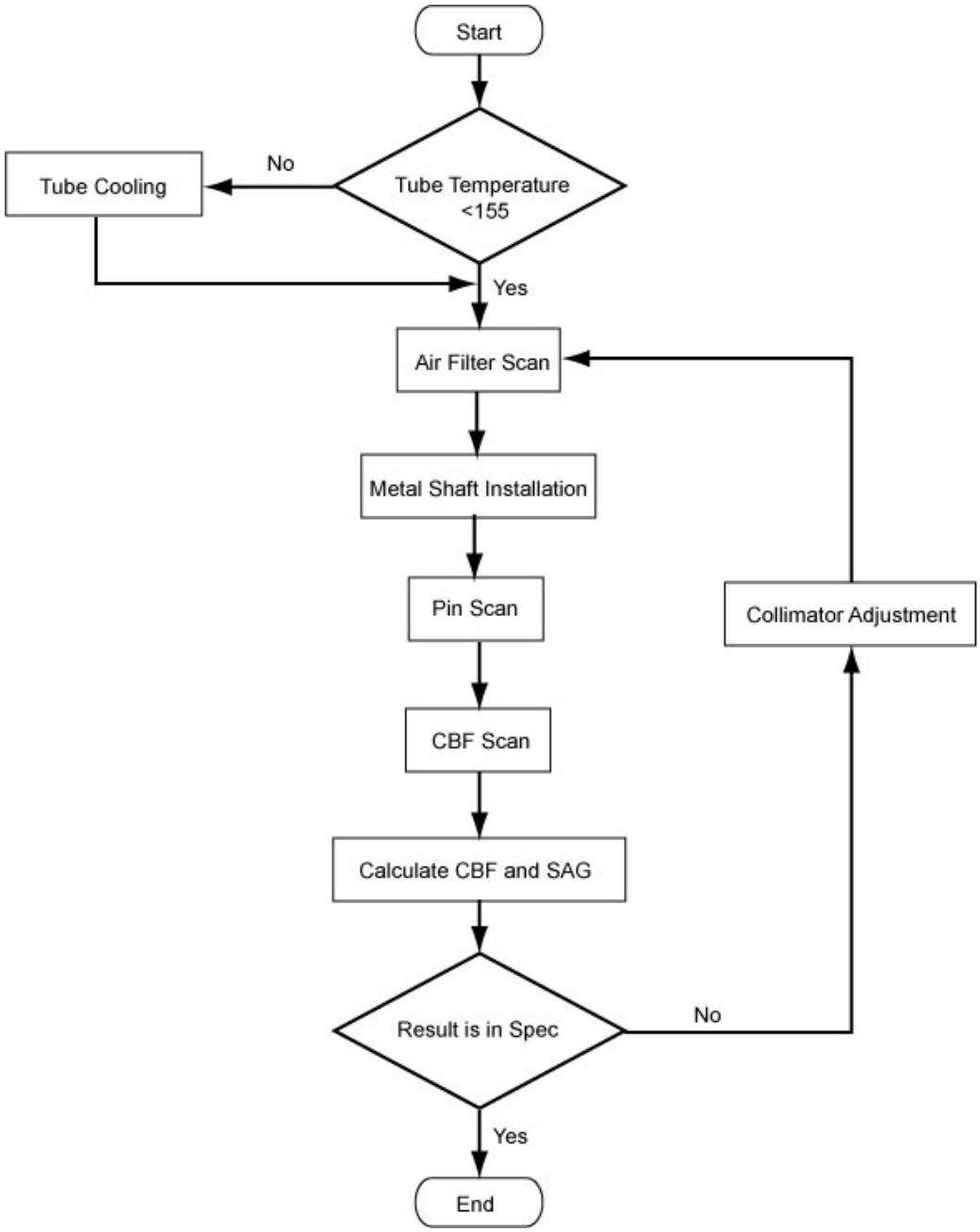
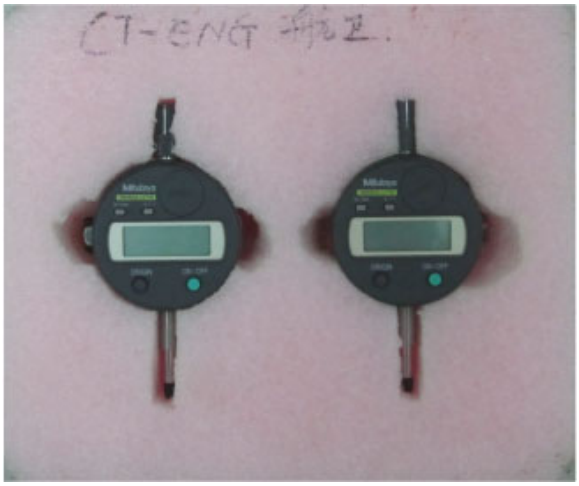


Illustration 2: Dial Gauge Indicator



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Dial Gauge Indicator (millimeters or inches) - see Illustration 2	1	-	-	-

3.2 Safety

⚠ CAUTION

No personnel or patients should be in the area of the gantry while scanning without adequate radiation safety precautions being utilized.

3.3 Required Conditions

Condition	Reference	Effectivity
Recommend procedure completion by GE Service trained personnel.	-	-
The positioning lights must be OK before CBF alignment.	-	-

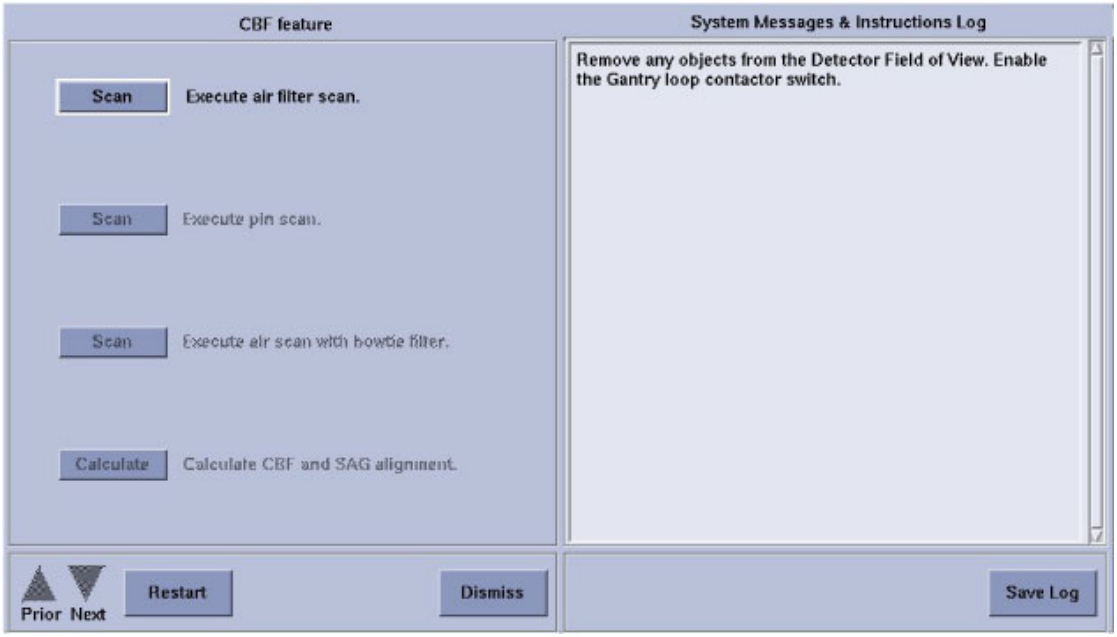
4 Procedure

4.1 Preparation

1. Move the cradle OUT to its fully retracted position (farthest from gantry) and position the table to its lowest height.
2. Remove the gantry left side cover.
3. Turn OFF “Axial Enable”, “HVDC” and “120VAC” switches on the service switch panel.
4. Remove the table foot switch cover.
5. Remove the gantry covers as required.
6. Turn ON “HVDC”, “Axial Enable” and “120VAC” switches.

4.2 Adjustment Procedure

1. Select [Service] -> [Calibration] -> [CBF and SAG Alignment] to enter ‘CBF feature’ UI.
2. Click [Scan] to start **Execute air filter scan**.



3. Hint information ‘CBF/SAG air scan (large spot) was successful.’ displays on the screen.
4. Insert the phantom holder into the cradle and attach a round, 0.125 inch diameter metal shaft to it.
5. Align the localization light on the metal shaft by referring information displayed on the screen.

Illustration 3: Hint Information (Red)

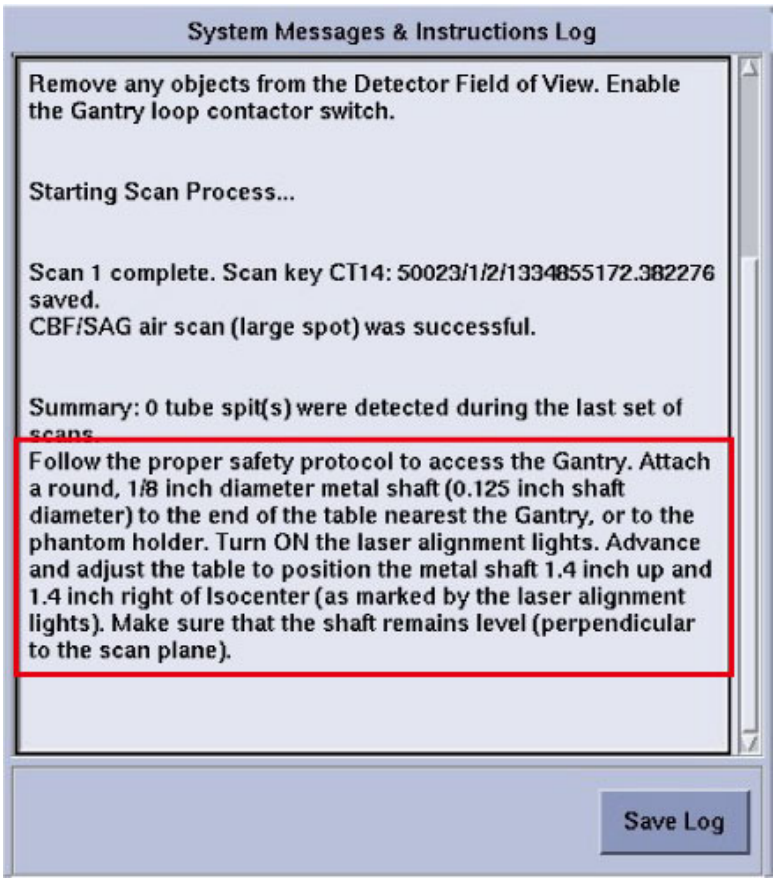
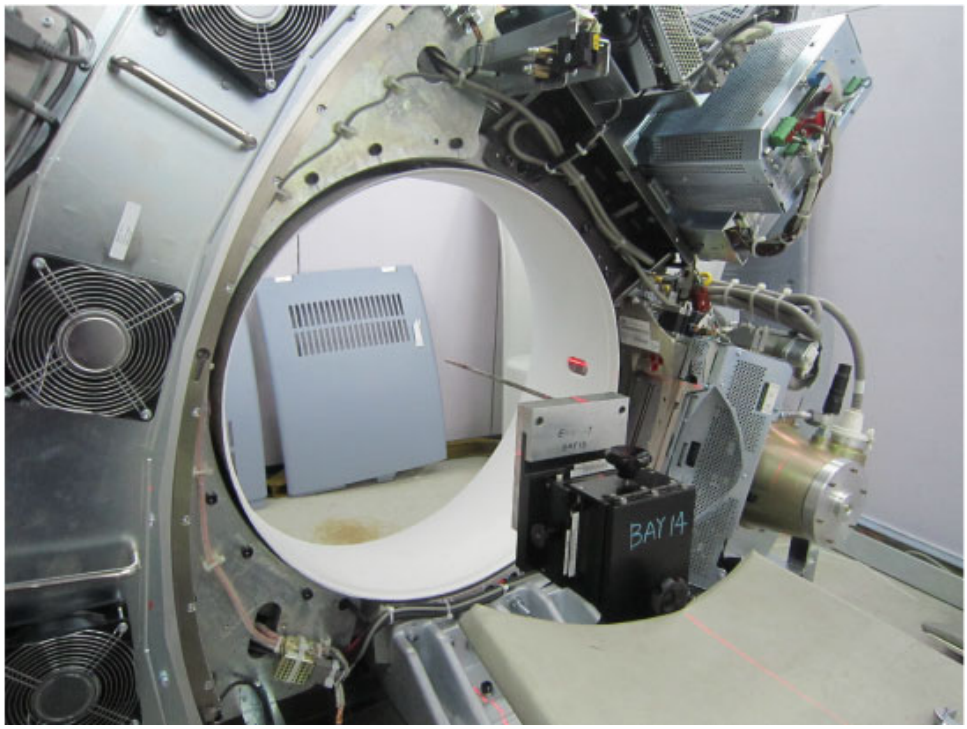


Illustration 4: Position Metal Shaft



6. Press [External Landmark] button to position the positioning light to the **190**.

Illustration 5: External Landmark



7. Press [Cradle In] button to move the cradle until **0** displays on the display/control panel.
8. Click [Scan] to start **Execute pin scan**.
9. Hint information 'CBF/SAG pin scan (large spot) was successful.' displays on the screen.
10. Move the metal shaft out of the detector field of view and click [Scan] to start **Execute air scan with bowtie filter**.
11. Click [Calculate] to start **Calculate CBF and SAG alignment**.
    a. If it is in specification, the following information displays:

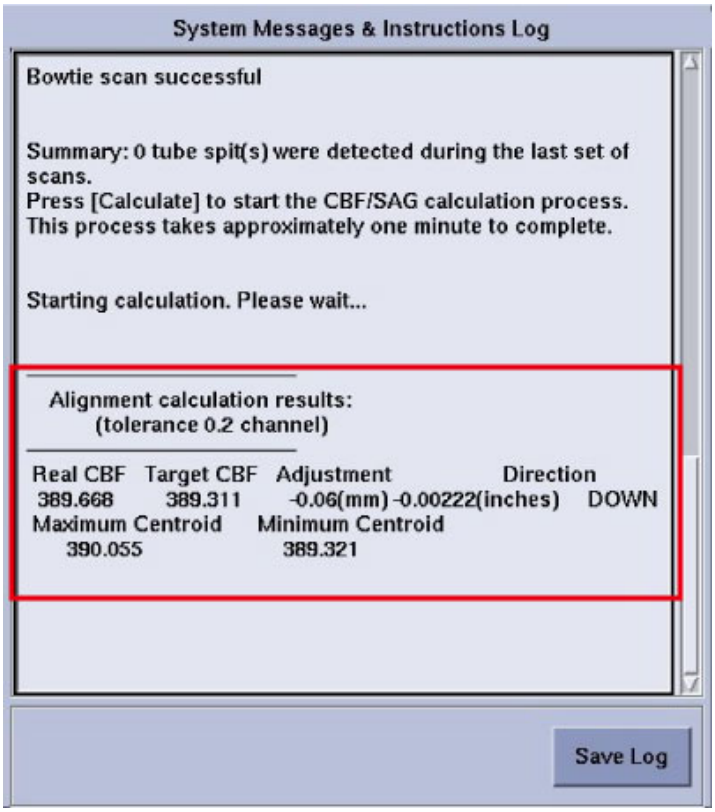
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The shift of collimator is xxxxx mm.

It is in specification: +/-xxxxx mm.

No need to adjust the position of the CBF/SAG.
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    b. If it is out of specification, the following information displays:

Illustration 6: Hint Information (Red)



12. Adjust the collimator position.

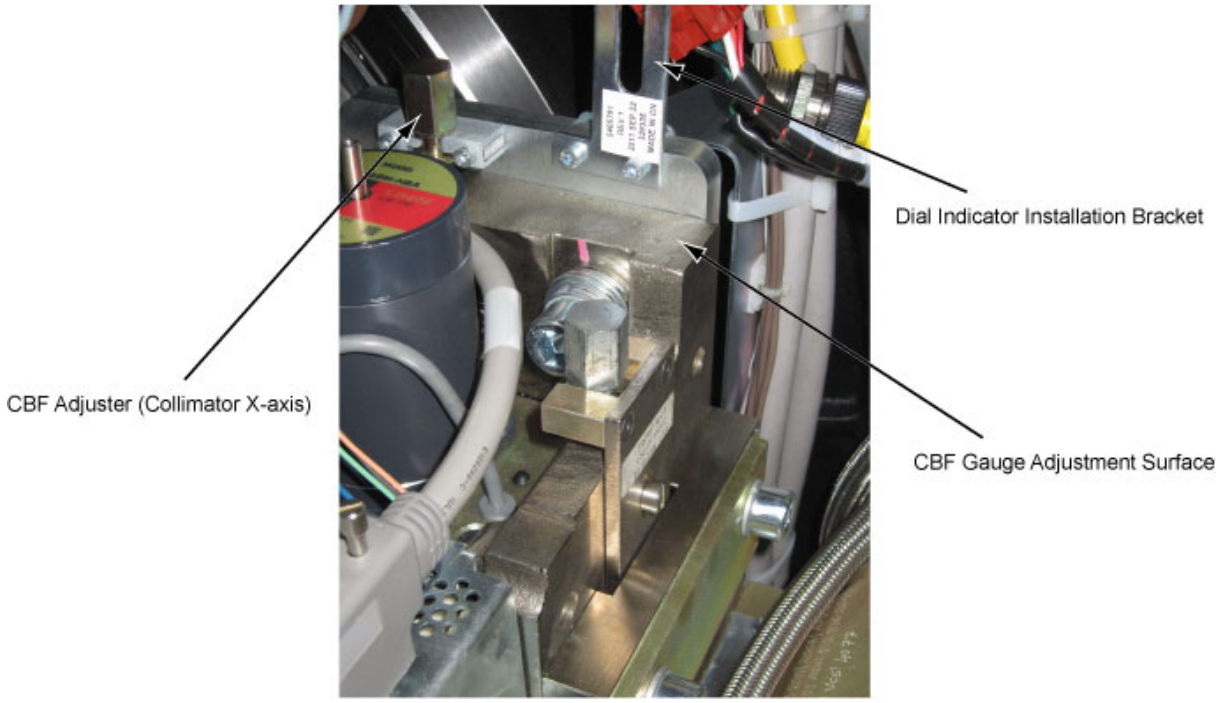
    a. Rotate the gantry until the tube reaches about 3 o'clock position.

    b. Turn OFF "HVDC", "Axial Enable" and "120VAC" switches on the service switch panel.

    c. Mount the dial indicator onto the collimator / tube assembly as shown below. Make sure you zero the dial indicator, when it is securely in place.

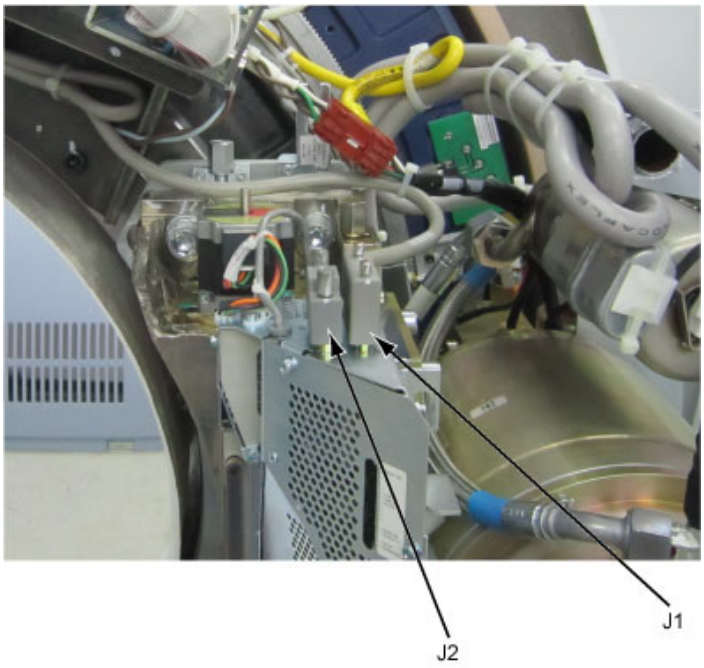
    NOTE: The dial indicator must be vertical with the CBF gauge adjustment surface.

Illustration 7: Dial Indicator Installation



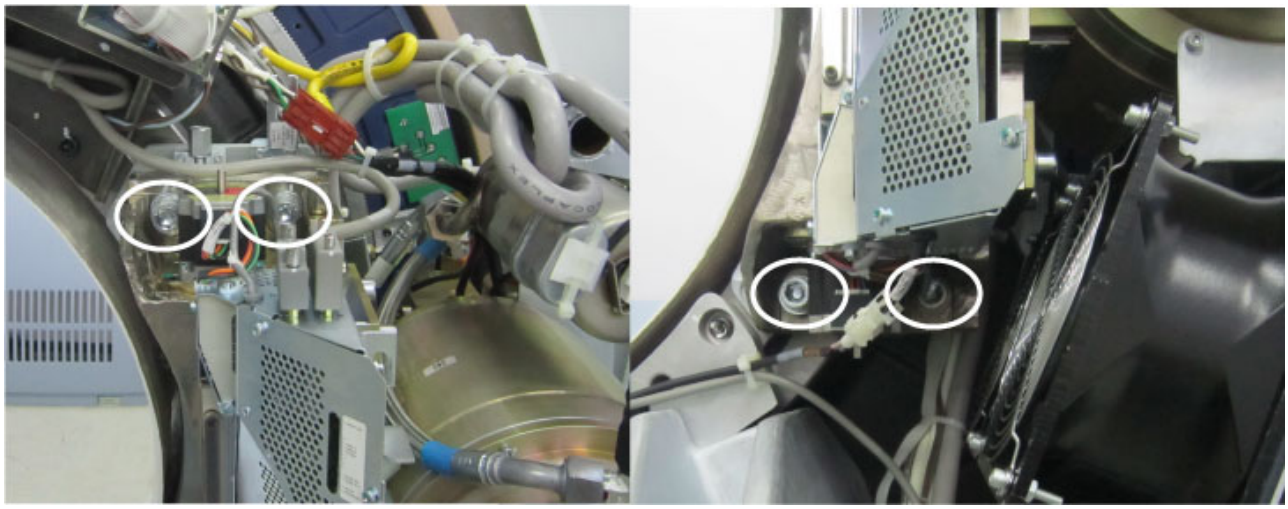
    d. Disconnect the cable connectors from the collimator assy J1 and J2 to be convenient for loosening screws of the collimator assy.

Illustration 8: Disconnect Cable Connectors from Collimator



e. Loosen the four M14 mounting bolts that secure the tube. (1/2 turn out is all that's necessary, and not any more.)

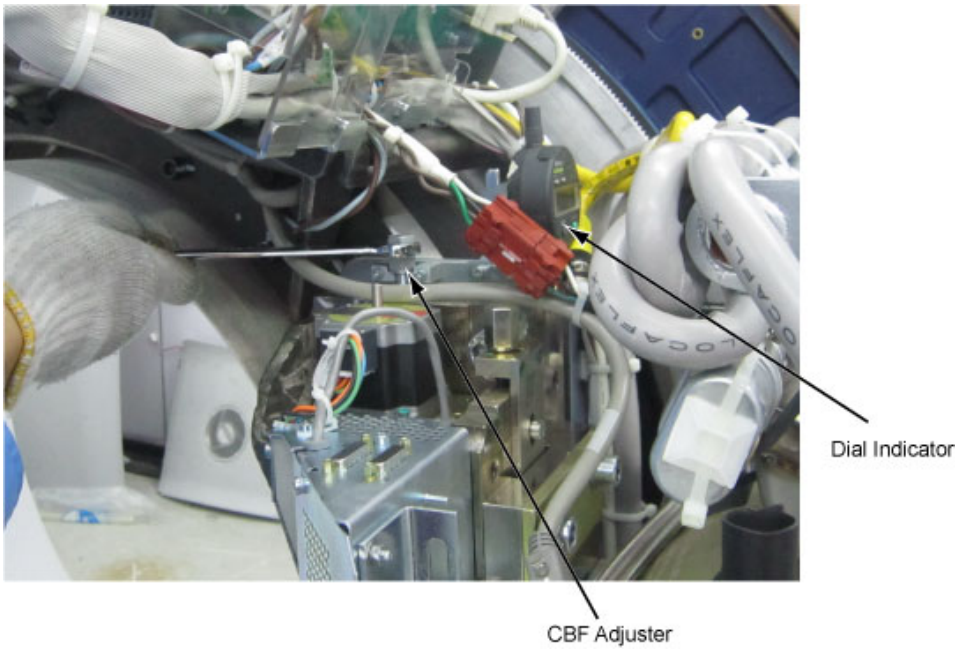
Illustration 9: Collimator Mounting Bolts



f. Adjust the collimator position in X-axis by turning the CBF adjuster according to the calculative result displayed on the monitor screen.

- If collimatorshift is positive, turn adjuster CW
- If collimatorshift is negative, turn adjuster CCW

Illustration 10: Adjust the CBF Adjuster



13. Secure M14 mounting bolts to load torque 96N-m.
14. Turn ON “Axial Enable”, “HVDC” and “120VAC” switches.
15. Click [Restart] to repeat CBF procedure from Step 2 to Step 11.
16. Click [Save Log], then click [Dismiss] to quit CBF.
17. Restore the gantry to original configuration and reinstall the table foot switch cover.

5 Finalization

No finalization steps.

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